SANA: LakeQA SUBSET20B results

Experiment records `2026-08-31-web-arm-subset20b` and `2026-08-31-model-tiers-subset20b`

1 Results

1.1 Retrieval source: open web versus the data lake

To test whether the retrieval axis is specific to the data lake or reflects a more general property of the agent, we replaced lake search with open-web search. The arm uses the Parallel Search API and, under `--no-s3`, loses every lake-backed file tool: the agent retains only `download`, which fetches an arbitrary `http(s)` URL into its sandbox, and `execute_code`. All four arms hold the remaining axes fixed at `search_results=naive`, `profile=standard` and `computation_tool=standard`, since the ideal computation tools resolve against authored data-lake records and would make a web run's outcome independent of what it retrieved.

Table 1 reports the result. The web arm reaches 35.0% semantic match against 45–50% for the three lake arms, and the three lake conditions are separated by a single task. The interesting quantity is not the gap in accuracy but the gap in retrieval: the ideal arm reaches 78.1% of the gold sources using 22.2 cycles, while the standard arm reaches 67.9% using 29.2 cycles and 56% more money. Oracle retrieval is markedly more efficient and yet converts to the same number of correct answers, which locates the bottleneck downstream of retrieval on this split.

Two properties of the web arm complicate its interpretation and we report them rather than correct for them. First, `download` accepts any URL, and the model routinely constructs one from pretrained knowledge rather than following a search result — live Socrata queries against `data.cityofnewyork.us`, `data.va.gov` export endpoints, and in one case a timestamped `web.archive.org` URL. The arm therefore measures an agent with open web access and code execution, not web-search retrieval, and its data path may be *stronger* than the lake arms', since it aggregates live authoritative sources rather than a fixed snapshot. Second, that liveness produces false negatives: two failures are arithmetically correct against current data but disagree with gold answers derived from the snapshot (2,700,968 against a gold 2,705,988; 2923.59 against 2941).

A third of all task-runs (33 of 80) reached the 30-call tool budget, and every arm scored roughly two to three times better below the cap than at it, so the absolute numbers here should be read as depressed by the budget rather than as ceilings.

Table 1: Retrieval-source comparison on LakeQA SUBSET20B (`gpt-5-mini`, 20 tasks per arm, zero errored runs). SM is semantic match (judge `gpt-5.4`); EM is lexical exact match. *Blank* counts answers the judge classified as `answer_unknown_blank`, i.e. honest abstentions rather than wrong answers. *Src* is the fraction of required gold sources the agent read; the web arm touches no lake sources by construction. *Cap* counts tasks that reached the 30-call tool budget.

Arm	SM (%)	EM (%)	Blank	Cycles	Tools	Cap	Cost ($)	Src (%)
Ideal	**50.0**	45.0	0	22.2	21.8	5	0.60	**78.1**
Standard	**50.0**	45.0	0	29.2	28.2	11	0.94	67.9
Naive	45.0	45.0	0	27.8	26.8	9	0.83	70.1
Web	35.0	30.0	1	26.2	25.2	8	1.00	—

Semantic match versus exact match. The two metrics differ by exactly one task, and that task is instructive. For the question whose gold answer is *"the Cut Bank Penguin"*, three arms answered `[Cut Bank`

`Penguin]` and the fourth answered `[World's largest statue…]`. All four are correct; all four score zero under exact match, because the lexical metric penalises a dropped article. On LakeQA most gold answers are a single token, where token-overlap F1 is identical to exact match by construction, and where it does differ it largely measures whether the model reproduced determiners. We therefore report semantic match throughout and do not report F1.

1.2 Replicate variance on the ablation grid

We ran the leave-one-out ablation over three model tiers on SUBSET20B, a 20-task split of LakeQA stratified jointly on node count and historical solve rate. Every axis is held at the oracle and one axis is degraded per cell, giving seven cells per model. To establish whether the resulting drops are measurable at this sample size, the two smaller tiers were run a second time under identical configuration.

The replicate is the central result. Across the 14 paired cells the mean absolute difference between runs is **7.1 pp** (median 5.0 pp, maximum 20.0 pp). Since a single task is worth 5 pp at $n = 20$, this places the resolution of the design near 15–20 pp rather than the 9–16 pp range the ablation was expected to produce. Table 2 reports both runs; Figure 1 plots each effect against the noise floor.

Three consequences follow. First, the apparent advantage of PRELOADED over the all-ideal reference (+10 pp at `gpt-5.4-nano` and +15 pp at `gpt-5-mini` on run 1 alone) does not survive replication, falling to +5.0 and +7.5 pp respectively. Second, one cell moved 20 pp between identical runs (`gpt-5.4-nano`, PLAN: NO PLAN, 25% → 45%), so no single-run reading of that condition is interpretable. Third, only two effects clear the floor by a comfortable margin: the search axis at −20 pp for BM25 at `gpt-5.4-nano` and −20 pp for PNEUMA at `gpt-5-mini`.

`gpt-5.2` was run once and therefore carries no error bar. Its search axis is notably flat (−5 pp for BM25, against −10 to −20 pp for the smaller tiers), which would indicate that retrieval quality matters less as model capability rises; we report it as an observation rather than a finding, since a single run at $n = 20$ cannot separate it from the variance measured on the other two tiers.

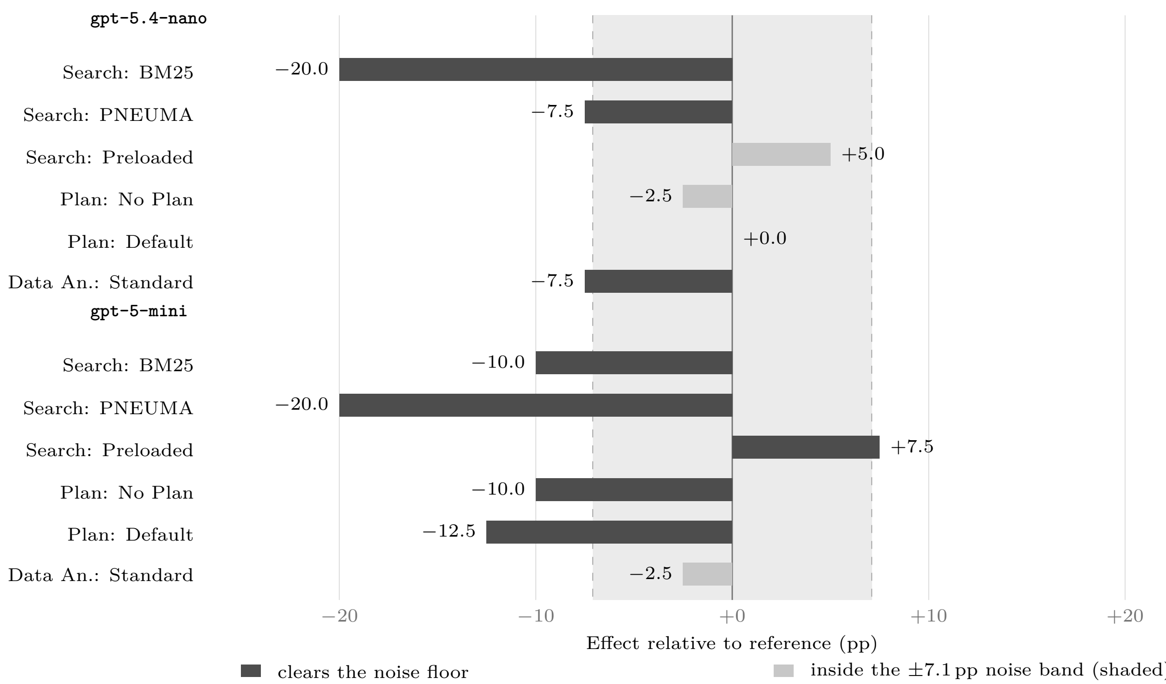

Figure 1: Ablation effects for the two replicated tiers. Each bar is the change from that model's all-ideal reference when one axis is degraded; the shaded column is the ±7.1 pp replicate noise floor measured across the 14 paired cells. Bars ending inside the shaded column cannot be distinguished from run-to-run variance at $n = 20$, where one task is worth 5 pp. `gpt-5.2` is omitted, having no replicate.

Table 2: Leave-one-out ablation across model tiers on LakeQA SUBSET20B (20 tasks per cell, exact match). R1 and R2 are two runs under identical configuration; $|\Delta|$ is their absolute difference and estimates the measurement error. $\bar{x}$ is their mean, and δ the effect relative to that model's reference cell. Effects exceeding the 7.1 pp replicate noise floor are set in bold. `gpt-5.2` was run once (†) and its effects therefore have no error bar.

Model	Condition	R1 (%)	R2 (%)	$\lvert\Delta\rvert$	$\bar{x}$ (%)	δ (pp)
`gpt-5.4-nano`	Reference (all ideal)	35	40	5	37.5	—
	Search: BM25	20	15	5	17.5	−20.0
	Search: PNEUMA	30	30	0	30.0	−7.5
	Search: Preloaded	45	40	5	42.5	+5.0
	Plan: No Plan	25	45	20	35.0	−2.5
	Plan: Default	40	35	5	37.5	0.0
	Data An.: Standard	35	25	10	30.0	−7.5
`gpt-5-mini`	Reference (all ideal)	65	70	5	67.5	—
	Search: BM25	50	65	15	57.5	−10.0
	Search: PNEUMA	45	50	5	47.5	−20.0
	Search: Preloaded	80	70	10	75.0	+7.5
	Plan: No Plan	60	55	5	57.5	−10.0
	Plan: Default	55	55	0	55.0	−12.5
	Data An.: Standard	60	70	10	65.0	−2.5
`gpt-5.2`†	Reference (all ideal)	60	—	—	60.0	—
	Search: BM25	55	—	—	55.0	−5.0
	Search: PNEUMA	60	—	—	60.0	0.0
	Search: Preloaded	80	—	—	80.0	+20.0
	Plan: No Plan	70	—	—	70.0	+10.0
	Plan: Default	55	—	—	55.0	−5.0
	Data An.: Standard	65	—	—	65.0	+5.0

Table 3: Replicate noise floor, computed over the 14 cells run twice (`gpt-5.4-nano` and `gpt-5-mini`, seven cells each). One task is worth 5 pp at $n = 20$, so the median disagreement between identical runs is a single task and the maximum is four.

Statistic over $\lvert R1 - R2\rvert$	Value (pp)
Mean	7.1
Median	5.0
Maximum	20.0
Paired cells	14